

GATE QUESTION ECE 2009 Q36

Question

Q.36

If $X = 1$ in the logic equation

$$[X + Z \{ \bar{Y} + (\bar{Z} + X\bar{Y}) \}] [\bar{X} + \bar{Z}(X + Y)] = 1,$$

then

(A) $Y = Z$

(B) $Y = \bar{Z}$

(C) $Z = 1$

(D) $Z = 0$

Answer: (C) $Z = 1$

Explanation:

Given $X = 1$, the expression simplifies as follows:

$$[1 + Z(\bar{Y} + (\bar{Z} + \bar{Y}))][0 + \bar{Z}(1 + Y)] = 1$$

Simplify the inner terms:

$$\bar{Y} + (\bar{Z} + \bar{Y}) = \bar{Y} + \bar{Z}$$

So,

$$[1 + Z(\bar{Y} + \bar{Z})][\bar{Z}(1 + Y)] = 1$$

The first term is always 1 (since $1 + \text{anything} = 1$). So we only need:

$$\bar{Z}(1 + Y) = 1$$

For this to be 1, $\bar{Z} = 1$ which means $Z = 0$. But wait—since $\bar{Z}(1 + Y)$ is supposed to be 1, let's analyze carefully:

- $\bar{Z} = 1 \implies Z = 0$ - Then $(1 + Y)$ must be 1 (always true), so $\bar{Z}(1 + Y) = 1$.

However, the given answer (C) says $Z = 1$. Let's double-check:

- If $Z = 1$: - Then $\bar{Z} = 0$. - So $\bar{Z}(1 + Y) = 0$. - Then the second term = 0. - But that would make the entire product 0, contradicting the equation.

If $Z = 0$: - Then $\bar{Z} = 1$. - Then $\bar{Z}(1 + Y) = 1$. - Then the product is $1 \cdot 1 = 1$.

Hence the correct answer is:

$(D) \ Z = 0$

Thus, the answer is actually (D).

Corrected Answer: (D) $Z = 0$